Kink Solitons and Kink-Antikink Collisions in the ϕ^4 Model

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MATH3001 Project in Mathematics
19 April 2024

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Contents

1 Introduction

The sources used in this section are [3], [6], [7] and [8].

Solitons are spatially localised wave-like solutions to certain non-linear partial differential equations. In the purest sense of the term, solitons are waves that keep a constant velocity and shape when travelling in a closed environment, and collide elastically with other waves of the same nature. However, many models exist whose solutions do not have these characteristics, but we still call them solitons if they have localised energy and maintain stability against small fluctuations [8]. Solitons frequently arise in various physical systems, so a better understanding of solitons means a better understanding of the world around us. Furthermore, the movement and interactions of solitons gives a very simple model of particle physics, exhibiting key characteristics such as the existence of antiparticles and the conservation of energy and momentum [6]. This makes the study of solitons an interesting and important area of research.

In this report, I will first introduce the non-linear Klein-Gordon Equation, and through doing so go over some important definitions and terminology to set up the foundational understanding of the maths behind kink solitons. The non-linear Klein-Gordon Equation gives rise to the sine-Gordon Equation and the ϕ^4 model, which will both also be presented. Next, I will cover the energy and momentum of kink solitons to shed more light on their characteristics, and the Bogomol'nyi argument to find a bound on the energy of a kink. The Lorentz boost will then be introduced, and I will use it when defining the initial conditions for the simulation of kink-antikink collisions. I will be numerically simulating the collisions for the ϕ^4 model, and the data that I found will be presented as part of the report. Finally, I will use my simulation to observe what happens when varying the initial velocity of the ϕ^4 kink-antikink pair, such as reflecting off each other, trapping into an oscillatory state, and various "bounce"-windows.

The code I made can be found on GitHub: https://github.com/moritzstuebing/Kink-Antikink-Collisions-in-the-Phi-4-Model-with-a-Finite-Difference-Method

1.1 A Brief History of Solitons

The first person to observe a soliton was Scotsman John Scott Russell in 1834, who noticed the wave that was created when a boat that was being dragged along a canal by two horses either side of it, came to a sudden stop. He noticed that the water that was being pushed along by the boat's movement carried on travelling along the canal, and noticed that it travelled with a smooth, unchanged shape and at a constant speed. He called this the wave of translation, which intuitively arises from the constant velocity and shape in the motion of the wave, making it appear as if the wave was simply translated from one spatial position to the next. [3][7]

Russell performed many experiments observing these waves, noting many of the properties that make them interesting to study today. However, it took a while for the mathematics to catch up to explain his discovery. It was not until 1895 that an equation admitting soliton solutions was found by Diederik Korteweg and Gustav de Vries. This is now known as the Korteweg-de Vries equation:

$$u_t + 6uu_x + u_{xxx} = 0 \tag{1}$$

[7]

This is a non-linear partial differential equation, where the solutions describe the movement of waves with long wavelengths in a shallow environment.

1.2 Ethical Considerations

While researching and creating the programme that simulates the kink-antikink collisions, it may have been tempting to falsify the results I observe, or exclusively take code from online sources to create the code itself. It would have been wrong to do this as then the code would not actually represent my own ability, and I would be presenting other people's work as my own. This programme and work has been researched from various papers, different examples of finite-difference schemes for other equations that I have used in part to create my own scheme as well as original briefing notes, and these have been referenced at the end. Any data I have used in my report, such as the values in the matrix U, has been found through the use of the numerical method outlined later on. It may have been tempting to find some other way of filling in the matrix, such as by manually filling it in for only times I choose to, to give the impression of a kink-antikink interaction. It would be wrong to do this as it would mean that I am presenting falsified data that is not accurate. All matrix values, and hence all plots that you see in the report, have come from the finite-difference scheme that I have developed.

2 Non-Linear Klein-Gordon Equation

Unless otherwise stated, everything in this section comes from [6], but proofs and solutions are my own work.

Another prominent partial differential equation that admits soliton solutions is the non-linear Klein-Gordon equation. This is the equation which we have studied in detail and gives rise to all kink solitons and systems in this report.

Definition 2.0.1 (Non-Linear Klein-Gordon Equation)**.** *Let $V : \mathbb{R} \to \mathbb{R}$ be a smooth function, with $V(u) \geq 0$ and $V(u) = 0$ at a discrete set of points. Then,*

$$u_{tt} - u_{xx} + V'(u) = 0 \tag{2}$$

is the non-linear Klein-Gordon equation.

The soliton solutions $u : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ are wave-like functions that depend on position x and time t. u_{tt} and u_{xx} are the second order partial derivatives of u with respect to t and x.

This gives us a general equation, but if we want to find solutions we need to specify what $V(u)$ is. Different choices for $V(u)$ can lead to very different behaviour in the generated systems.

Definition 2.0.2 (Potential)**.** *A smooth function $V : \mathbb{R} \to \mathbb{R}$ that satisfies (2), such that $V(u) \geq 0$ and $V(u) = 0$ at a discrete set of points, is called the potential of the equation.*

As mentioned, different potentials give rise to differing equations and models, which lead to different solutions. When $V(u) = 1 - \cos(u)$, we call (2) the sine-Gordon equation, and when $V(u) = \frac{1}{2}(1 - u^2)^2$ we call (2) the ϕ^4 equation or model. The solutions to these equations exhibit contrasting behaviour, especially in collisions between solutions. These solutions will be covered in some more detail later on.

Definition 2.0.3 (Vacuum of the Potential)**.** *Any value $u_i \in \mathbb{R}$ such that $V(u_i) = 0$ is called a vacuum of the potential V(u).*

For example, the ϕ^4 model has two vacua: $x = 1$ and $x = -1$.

Proposition 2.0.4 (Vacuum Solutions)**.** *Any constant field $u(x, t) = c$ where c is some vacuum of the potential V, is a solution of the Klein-Gordon equation (2).*

Proof. Let $u(x, t) = c$, where c is a vacuum of V, so $V(c) = 0$. As c is a constant, we know that $u_{tt} = u_{xx} = 0$. Furthermore, as $V(u) \geq 0$ for all $u \in \mathbb{R}$, and $V(c) = 0$, then c must be a minimum point of the potential V. Hence, as V attains a minimum at c, its derivative is $V'(c) = 0$

Substituting our values for u_{tt}, u_{xx} and V' into equation 2, we get $0 - 0 + 0 = 0$ so the constant field $u(x, t) = c$ is a solution of the Klein-Gordon equation. $\square$

2.1 Kink Solitons

The boundary behaviour for kinks is dependent on the vacua of the specified potential. So, now that we have defined a vacuum of the potential, we are in a position to define what kink soliton solutions of the equation (2) are.

Definition 2.1.1 (Kinks/Antikinks)**.** *Given a potential V that has more than one vacuum point, where u_0, u_1 are adjacent vacua of the potential with $u_0 < u_1$, then a kink soliton of* *the non-linear Klein-Gordon equation is any solution $u(x,t)$ of (2) that obeys the following boundary behaviour:*

$$\lim_{x\to-\infty} u(x,t) = u_0$$
$$\lim_{x\to\infty} u(x,t) = u_1$$

We call solutions that obey the opposite boundary behaviour

$$\lim_{x\to-\infty} u(x,t) = u_1$$
$$\lim_{x\to\infty} u(x,t) = u_0$$

antikinks.

At the x-boundaries of our solutions, the kinks tend to two of the vacua of the potential. As x tends to $+/-\infty$, the boundary behaviour of an antikink is the opposite of the corresponding kink. Hence, the shape of an antikink can be understood as the reflection of the corresponding kink in the vertical axis.

Proposition 2.1.2. *Let $u_K : \mathbb{R} \to \mathbb{R}$ be a kink solution of (2). Then $u_{\tilde{K}} : \mathbb{R} \to \mathbb{R}$ such that*

$$u_{\tilde{K}}(x,t) = u_K(-x,t)$$

is an antikink solution of (2).

Proof. Let $u_{\tilde{K}}(x,t)$ be defined as above. Let us work out $u_{\tilde{K}_{tt}}(x,t)$ and $u_{\tilde{K}_{xx}}(x,t)$. Since t remains the same, we have that $u_{\tilde{K}_{tt}}(x,t) = u_{K_{tt}}(-x,t)$. Using the chain rule, we work out that $u_{\tilde{K}_{xx}}(x,t) = u_{K_{xx}}(-x,t)$

Hence, we have that

$$u_{\tilde{K}_{tt}}(x,t) - u_{\tilde{K}_{xx}}(x,t) + V'(u_{\tilde{K}}) = u_{K_{tt}}(-x,t) - u_{K_{xx}}(-x,t) + V'(u_K) = 0$$

since u_K is a solution of (2). So, $u_{\tilde{K}}$ is a solution to (2).

Furthermore, since x is negative in $u_{\tilde{K}}$, and $u_K(x,t)$ obeys kink boundary behaviour, the boundary behaviour for $u_{\tilde{K}}$ is the opposite of $u_K(x,t)$ in the x-direction. Explicitly, as $x \to -\infty$, $u_{\tilde{K}} \to u_1$ and as $x \to \infty$, $u_{\tilde{K}} \to u_0$. So, $u_{\tilde{K}}$ obeys antikink boundary behaviour and is a solution of (2). Hence, it is an antikink solution of (2). □

Everything necessary to understand some different models and solutions has now been introduced. Let us analyse two specific potentials that give rise to different systems.

2.2 Sine-Gordon Equation

As mentioned earlier, when the potential $V(u) = 1 - \cos u$ (so $V'(u) = \sin u$) we call (2) the sine-Gordon equation:

$$u_{tt} - u_{xx} + \sin u = 0 \tag{3}$$

Figure 1 shows a graph of the Sine-Gordon potential $V(u) = 1 - \cos(u)$ for $-2\pi \leq u \leq 2\pi$. If the x-axis stretched off to positive and negative infinity, then we would observe that there are vacua at $u = 2\pi n$ for all $n \in \mathbb{N}$. Since kink/antikink solutions depend on having boundary behaviour corresponding to vacua, and there is an infinite set of vacua in the sine-Gordon equation, there is also an infinite number of kink solutions to the sine-Gordon equation.

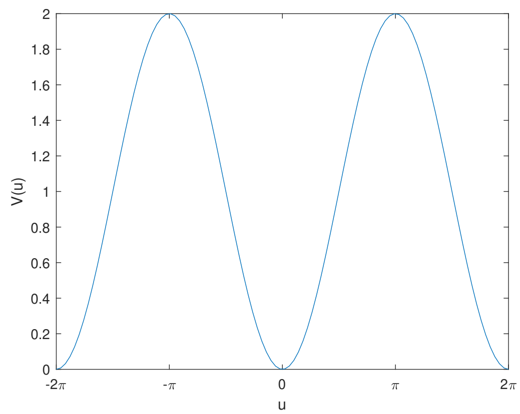

Figure 1: Graph of $V(u) = 1 - \cos(u)$, for $-2\pi \leq u \leq 2\pi$

Clearly, $V(u) \geq 0$ for all $u \in \mathbb{R}$, and only takes the value $V(u) = 0$ at points that are a multiple of 2π (a discrete set of points), so the conditions for a potential V are satisfied.

One particular solution to (3) is

$$u(x,t) = 4 \tan^{-1} e^{\gamma(x-vt)} \tag{4}$$

where $\gamma = (1 - v^2)^{-\frac{1}{2}}$ and $v \in (-1, 1)$ is the velocity of the kink.

Let me show that this $u(x,t)$ is a kink solution of (3), which I will do by showing that it satisfies the equation, and obeys the necessary boundary behaviour.

Let $u(x,t)$ be defined as in (4).

We have

$$u_{xx} = \frac{4\gamma^2(e^{\gamma(x-vt)} - e^{3\gamma(x-vt)})}{(e^{2\gamma(x-vt)} + 1)^2}$$

$$u_{tt} = -\frac{4\gamma^2 v^2(e^{3\gamma(x-vt)} - e^{\gamma(x-vt)})}{(e^{2\gamma(x-vt)} + 1)^2}$$

$$\sin(u) = V'(u) = \frac{4(e^{\gamma(x-vt)} - e^{3\gamma(x-vt)})}{(e^{2\gamma(x-vt)} + 1)^2}$$

Substituting our expressions for u_{xx}, u_{tt}, and $V'(u)$ into (2), we get

$$-\frac{4\gamma^2 v^2(e^{3\gamma(x-vt)} - e^{\gamma(x-vt)})}{(e^{2\gamma(x-vt)} + 1)^2} - \frac{4\gamma^2(e^{\gamma(x-vt)} - e^{3\gamma(x-vt)})}{(e^{2\gamma(x-vt)} + 1)^2} + \frac{4(e^{\gamma(x-vt)} - e^{3\gamma(x-vt)})}{(e^{2\gamma(x-vt)} + 1)^2}$$

$$= \frac{4\gamma^2 v^2(e^{\gamma(x-vt)} - e^{3\gamma(x-vt)})}{(e^{2\gamma(x-vt)} + 1)^2} - \frac{4\gamma^2(e^{\gamma(x-vt)} - e^{3\gamma(x-vt)})}{(e^{2\gamma(x-vt)} + 1)^2} + \frac{4(e^{\gamma(x-vt)} - e^{3\gamma(x-vt)})}{(e^{2\gamma(x-vt)} + 1)^2}$$

Dividing by the common term, we arrive at the expression

$$\begin{aligned} &4\gamma^2 v^2 - 4\gamma^2 + 4 \\ &= \gamma^2 v^2 - \gamma^2 + 1 \\ &= -\gamma^2(1 - v^2) + 1 \\ &= -1 + 1 = 0 \end{aligned}$$

as $\gamma = (1 - v^2)^{-\frac{1}{2}}$.

As shown, when substituting $u(x, t)$ into (3), you arrive at $0 = 0$, so $u(x, t)$ is a solution of (3).

$V(u) = 1 - \cos u = 0$ when $u_0 = 0$ and when $u_1 = 2\pi$. Note that these are not the only vacua of the potential, but that they are indeed neighbouring vacua with $u_0 < u_1$.

We have

$$u_0 = \lim_{x \to -\infty} u(x, t) = \lim_{x \to -\infty} 4 \tan^{-1} e^{\gamma(x - vt)} = 4 \times 0 = 0$$

and

$$u_1 = \lim_{x \to \infty} u(x, t) = \lim_{x \to \infty} 4 \tan^{-1} e^{\gamma(x - vt)} = 4 \times \frac{\pi}{2} = 2\pi$$

Hence, $u(x, t)$ satisfies the boundary conditions for a kink, i.e it tends to a vacuum u_1 of the potential as x increases to ∞, and it tends to a vacuum u_0 of the potential as x decreases to $-\infty$, where $u_0 < u_1$

So, (4) is a kink solution to (3).

2.3 The ϕ^4 Model

Another equation that admits soliton solutions is the ϕ^4 equation. This model arises when the potential is $V(u) = \frac{1}{2}(1 - u^2)^2$, giving the following equation:

$$u_{tt} - u_{xx} - 2u(1 - u^2) = 0 \tag{5}$$

As there are only two vacua for this potential (at $x = 1$ and $x = -1$), there is only one kink solution and one antikink solution. This means that the only soliton collisions that can be studied in this model are kink-antikink collisions, which will be investigated in detail later on.

The kink solution for the ϕ^4 model is as follows:

$$u(x, t) = \tanh(\gamma(x - vt)) \tag{6}$$

and the corresponding antikink solution is:

$$u(x, t) = -\tanh(\gamma(x - vt)) \tag{7}$$

Again, $\gamma = (1 - v^2)^{-\frac{1}{2}}$, where $v \in (-1, 1)$ is the velocity of the kink/antikink.

Figure 2 shows the ϕ^4 potential. The vacua at $x = 1$ and $x = -1$ can be seen on this plot. Figure 3 gives a plot of a ϕ^4 kink for two different times.

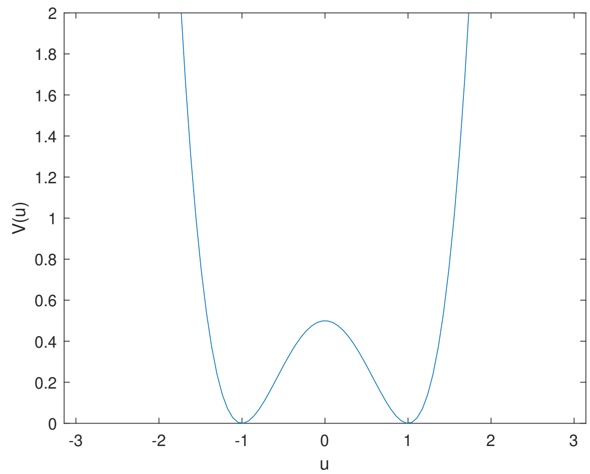

Figure 2: Graph of the ϕ^4 potential $V(u) = \frac{1}{2}(1-u^2)^2$

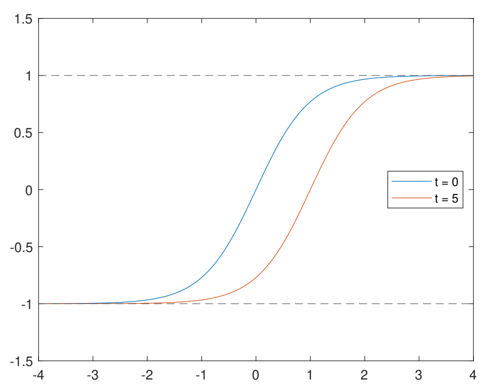

Figure 3: Graph of the solution (6) of the ϕ^4 model. Graphed for velocity $v = 0.2$ at times $t = 0$, $t = 5$.

Let me show that (6) is a kink solution of the ϕ^4 model.

Using the chain rule, and the hyperbolic trigonometric identity $\operatorname{sech}^2 x = 1 - \tanh^2 x$, we arrive at the following for the second order partial derivatives and expression for $V'(u)$:

$$\begin{aligned} u_{tt} &= -2\gamma^2 v^2 \tanh(\gamma(x - vt)) \operatorname{sech}^2(\gamma(x - vt)) \\ u_{xx} &= -2\gamma^2 \tanh(\gamma(x - vt)) \operatorname{sech}^2(\gamma(x - vt)) \\ V'(u) &= -2u(1 - u^2) = -2 \tanh(\gamma(x - vt)) \operatorname{sech}^2(\gamma(x - vt)) \end{aligned}$$

Plugging into our ϕ^4 equation, and dividing by the common term, we get

$$-2\gamma^2 v^2 + 2\gamma^2 - 2 = 2\gamma^2(1 - v^2) - 2$$

Since $\gamma = (1 - v^2)^{-\frac{1}{2}}$, so $\gamma^2 = \frac{1}{1-v^2}$ we have

$$2 - 2 = 0$$

Hence, (6) is a solution to the ϕ^4 model. Now I will show that it is a kink solution.

As mentioned, the vacua for this model are at $u_0 = -1$ and $u_1 = 1$. Clearly, $u_0 < u_1$. Hence, any kink solution of the model should tend to -1 as x tends to minus infinity, and to 1 as x tends to infinity. The antikink solution should exhibit the opposite boundary behaviour.

As x tends to minus infinity, (6) tends to -1, and as x tends to infinity, (6) tends to 1. This is due to the nature of tanh. So, (6) exhibits the correct kink boundary behavior for the ϕ^4 model. Hence, (6) is a kink solution of the ϕ^4 model.

3 Laws of Conservation

Unless otherwise stated, everything in this section comes from [6], but proofs and solutions are my own work.

If there are no external forces acting on a system, then the laws of conservation state that the momentum and energy of the system, will remain the same (be conserved) at all times. This also true for our kink solitons, and in this section we will develop a deeper understand of the momentum and energy of kink solitons.

Definition 3.0.1 (Energy and Momentum of a Field u)**.** *The energy and momentum of a field $u : \mathbb{R} \to \mathbb{R}$ can be defined as follows:*

$$E(t) = \int_{-\infty}^{\infty} (\frac{1}{2}{u_t}^2 + \frac{1}{2}{u_x}^2 + V(u))dx \tag{8}$$

$$P(t) = -\int_{-\infty}^{\infty} u_t u_x dx \tag{9}$$

Clearly, both the energy and the momentum of a field depend on t only. We can use these equations for $E(t)$ and $P(t)$ to work out the energy and momentum of a kink, however I will return to find the energy and momentum of (4) and (6) once we have covered the invariance of the Lorentz boost, as this makes the task much simpler.

Proposition 3.0.2. *If a solution of (2) has finite energy and momentum, then both $E(t)$ and $P(t)$ are constant.*

Proof. To show that $E(t)$ is constant, I will show that $\frac{dE}{dt} = 0$.

$$\begin{aligned}
\frac{dE}{dt} &= \int_{-\infty}^{\infty} (u_{tt}u_t + u_{xt}u_x + V'(u)u_t)dx \\
&= \int_{-\infty}^{\infty} (u_t(u_{tt} + V'(u)) + u_x u_{xt})dx \\
&= \int_{-\infty}^{\infty} (u_t u_{xx} + u_x u_{xt})dx \\
&= \int_{-\infty}^{\infty} \frac{\partial}{\partial x}(u_t u_x)dx \\
&= u_t u_x \Big|_{-\infty}^{\infty} = 0
\end{aligned}$$

This equals 0 because $u(x,t)$ tends to vacua (which are constant) at $x = +/-\infty$, which means that the partial derivatives of u at the boundaries are 0. Hence, $E(t)$ is constant.

$$\frac{dP}{dt} = -\int_{-\infty}^{\infty} (u_{tt}u_x + u_{xt}u_t)dx$$
$$= -\int_{-\infty}^{\infty} ((u_{xx} - V'(u))u_x + u_{xt}u_t)dx$$
$$= -\int_{-\infty}^{\infty} (u_x u_{xx} + u_{xt}u_t - V'(u)u_x)dx$$
$$= -(\frac{1}{2}u_x^2 + \frac{1}{2}u_t^2 - V(u))\Big|_{-\infty}^{\infty} = 0$$

This equals 0 by the same reasoning as in the $E(t)$ case, and because u tends to vacua of the potential at $x = +/-\infty$, meaning that $V(u) = 0$ at $x = +/-\infty$. Hence, $P(t)$ is constant.

□

3.1 Energy Density

Notice that equations (8) and (9) give the total energy or momentum of a field at a given time. If we want to find the distribution of the energy in a kink for a given time and velocity, we will need an equation for energy density. This equation is a function of x, so we can use it to graph the energy density with respect to x of a kink, giving us a picture of whereabouts in the kink the most/least energy lies.

Definition 3.1.1 (Energy Density)**.** *The energy density of the field u is*

$$\mathcal{E}(x) = \frac{1}{2}u_t^2 + \frac{1}{2}u_x^2 + V(u)$$

Let me find the energy density for the kink solution (6) of the ϕ^4 model.

Let $u(x,t) = \tanh(\gamma(x - vt))$. Using the hyperbolic trigonometric identity $1 - \tanh^2(x) = \operatorname{sech}^2(x)$, we can find the first-order space and time derivatives, as well as the potential $V(u)$, of $u(x,t)$:

$$u_t = -\gamma v(1 - \tanh^2(\gamma(x - vt)) = -\gamma v \operatorname{sech}^2(\gamma(x - vt))$$
$$u_x = \gamma(1 - \tanh^2(\gamma(x - vt)) = \gamma \operatorname{sech}^2(\gamma(x - vt))$$
$$V(u) = \frac{1}{2}(1 - \tanh^2(\gamma(x - vt))^2 = \frac{1}{2}\operatorname{sech}^4(\gamma(x - vt))$$

So, the energy density $\mathcal{E}(x)$ is

$$\mathcal{E}(x) = \frac{1}{2}\gamma^2 v^2 \operatorname{sech}^4(\gamma(x - vt)) + \frac{1}{2}\gamma^2 \operatorname{sech}^4(\gamma(x - vt)) + \frac{1}{2}\operatorname{sech}^4(\gamma(x - vt))$$
$$= \frac{1}{2}((\gamma^2(1 + v^2) + 1)\operatorname{sech}^4(\gamma(x - vt)))$$
$$= \gamma^2 \operatorname{sech}^4(\gamma(x - vt))$$

Figure 4 gives the energy density of two solutions to the ϕ^4 model, one of which is plotted for two different times. It shows that the largest concentration of energy is stored in the center of the kink, and moves corresponding to the movement of the kink over time. For greater initial velocities, the energy density of the kink is greater, as the figure depicts.

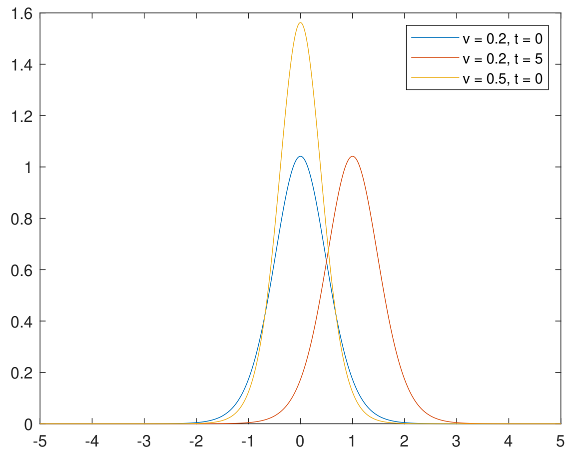

Figure 4: Graph of the energy density of solution (6) of the ϕ^4 model. Graphed for velocity $v = 0.2$ at times $t = 0$, $t = 5$, and graphed for velocity $v = 0.6$ at time $t = 0$.

4 The Bogomol'nyi Argument

Unless otherwise stated, everything in this section comes from [6], but proofs and solutions are my own work.

The Bogomol'nyi argument is used to find a bound on the energy of a kink. In order to apply it, we first need to study static kinks.

4.1 Static Kinks

Definition 4.1.1 (Static Kinks)**.** *Static kinks are kink solutions of (3) that do not depend on time t.*

As a result of this, all partial derivatives of a static kink u with respect to t equal 0.

Theorem 4.1.2. *u is a static kink if and only if it satisfies the first order ODE*

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \sqrt{2V(u)}$$

Proof. ($\rightarrow$) Let u be a static kink, so u doesn't depend on t. Consequently, this means that $u_{tt} = 0$, so (2) becomes $\frac{d^2u}{dx^2} = V'(u)$. Next, multiply this equation by $\frac{du}{dx}$:

$$\frac{du}{dx}\frac{d^2u}{dx^2} = \frac{du}{dx}V'(u) \leftrightarrow \frac{d}{dx}(\frac{1}{2}(\frac{du}{dx})^2) = \frac{d}{dx}(V(u))$$

Integrating both sides we get

$$(\frac{1}{2}(\frac{du}{dx})^2) = (V(u))$$

which implies that

$$\frac{du}{dx} = \sqrt{2V(u)}$$

The proof in the ($\leftarrow$) direction is analogous, the steps taken are just in the reverse order. I will omit this to save space. $\square$

In Section 6, I will be using a static kink and antikink of the ϕ^4 model to define the initial conditions of our numerical simulation for kink-antikink collisions.

4.2 The Bogomol'nyi Bound

Evgeny Bogomol'nyi developed an argument to show that the energy E of a static kink is bounded below by what we now call the Bogomol'nyi bound. As the argument has been presented in the general case in the briefing notes, I will use the argument to find a bound on the energy of a ϕ^4 kink, i.e where $V(u) = \frac{1}{2}(1-u^2)^2$.

Let $u(x,t)$ be a field with kink boundary behaviour. First, we need to find an increasing function $F(u)$ such that $V(u) = \frac{1}{2}F'(u)^2$. Well, this is very simple in this case, as we can just take $F'(u) = 1-u^2$. Then $F(u) = u - u^3$.

Then, using equation (8) for the energy of a kink, we have

$$E(t) = \int_{-\infty}^{\infty} (\frac{1}{2}{u_t}^2 + \frac{1}{2}{u_x}^2 + \frac{1}{2}(1-u^2)^2)dx$$

Since $\frac{1}{2}{u_t}^2$ is strictly positive, we can take out this term and bound the expression below by

$$\geq \int_{-\infty}^{\infty} (\frac{1}{2}{u_x}^2 + \frac{1}{2}(1-u^2)^2)dx$$

Now we can complete the square to get

$$= \frac{1}{2}\int_{-\infty}^{\infty} (u_x - (1-u^2))^2 dx + \int_{-\infty}^{\infty} u_x(1-u^2)dx$$

The former term is always positive as it is a square, so again we can take this term out of the expression, and bound the expression below by the following

$$\geq \int_{-\infty}^{\infty} u_x(1-u^2)dx$$

$$= (u - \frac{1}{3}u^3)\Big|_{-\infty}^{\infty}$$

$$= F(u)\Big|_{-\infty}^{\infty}$$

$$= F(1) - F(-1) = \frac{4}{3}$$

due to the boundary behaviour of $u(x,t)$. So,

$$E(t) \geq \frac{4}{3}$$

With equality if and only if $u(x,t)$ is a static kink (as then $u_t = 0$) and $u_x = 1-u^2$ (i.e $u_x = \sqrt{2V(u)}$). If this is the case, then the two inequalities in the argument above become equalities.

Theorem 4.2.1 (Bogomol'nyi Bound)**.** *Given a potential $V(u)$, the energy of a kink/antikink corresponding to this potential, with rest energy M, is bounded below by the following*

$$E(t) \geq M = \int_{u_0}^{u_1} \sqrt{2V(u)}du$$

with equality if and only if u is static and $u_x = \sqrt{2V(u)}$ for a kink and $u_x = -\sqrt{2V(u)}$ for an antikink.

5 Lorentz Boost and Invariance

Unless otherwise stated, everything in this section comes from [6], but proofs and solutions are my own work.

Lorentz boosting a static kink involves imparting it with a velocity that will allow it to move over time. In Section 6, I will be using the Lorentz boost of a static kink and antikink when defining the initial conditions of my finite difference scheme.

Let us define a new field $\tilde{u}$ from $u(x,t)$, where u is any field:

$$\tilde{u}(x,t) = u(\gamma(x - vt), \gamma(t - vx))$$

$v \in (-1, 1)$ and $\gamma = (1 - v^2)^{-\frac{1}{2}}$

Proposition 5.0.1. *$\tilde{u}$ is a solution of the Klein-Gordon Equation if and only if u is a solution of the Klein-Gordon Equation.*

Proof. Let $\tilde{u}(x,t) = u(\gamma(x - vt), \gamma(t - vx)) = u(\tilde{x}, \tilde{t})$. So, $\tilde{x} = \gamma(x - vt)$ and $\tilde{t} = \gamma(t - vx)$. Then, using the chain rule

$$u_{tt} = \gamma^2 v^2 \frac{\partial^2 u}{\partial \tilde{x}^2} - 2\gamma^2 v \frac{\partial u}{\partial \tilde{x} \partial \tilde{t}} + \gamma^2 \frac{\partial^2 u}{\partial \tilde{t}^2}$$

and

$$u_{xx} = \gamma^2 \frac{\partial^2 u}{\partial \tilde{x}^2} - 2\gamma^2 v \frac{\partial u}{\partial \tilde{x} \partial \tilde{t}} + \gamma^2 v^2 \frac{\partial^2 u}{\partial \tilde{t}^2}$$

Substituting these expressions for u_{tt} and u_{xx} into (2) we arrive at the following:

$$\begin{aligned} u_{tt} - u_{xx} + V'(u) &= \gamma^2 v^2 \frac{\partial^2 u}{\partial \tilde{x}^2} + \gamma^2 \frac{\partial^2 u}{\partial \tilde{t}^2} - \gamma^2 \frac{\partial^2 u}{\partial \tilde{x}^2} + \gamma^2 v^2 \frac{\partial^2 u}{\partial \tilde{t}^2} + V'(u) = 0 \\ &= \gamma^2 (1 - v^2) \frac{\partial^2 u}{\partial \tilde{t}^2} - \gamma^2 (1 - v^2) \frac{\partial^2 u}{\partial \tilde{x}^2} + V'(u) = 0 \end{aligned}$$

Since $\gamma = (1 - v^2)^{-\frac{1}{2}}$, we arrive at the following:

$$u_{\tilde{t}\tilde{t}} - u_{\tilde{x}\tilde{x}} + V'(u) = 0$$

Since $\tilde{u}(x,t) = u(\gamma(x - vt), \gamma(t - vx)) = u(\tilde{x}, \tilde{t})$, this is equivalent to

$$\tilde{u}_{tt} - \tilde{u}_{xx} + V'(\tilde{u}) = 0$$

Hence, $\tilde{u}$ is a solution to (2). □

Definition 5.0.2 (Lorentz Boost)**.** *The Lorentz boost $\tilde{u}$, with velocity v, of any field u is defined as the following:*

$$\tilde{u}(x,t) = u(\gamma(x - vt)) \tag{10}$$

where $\gamma = (1-v^2)^{-\frac{1}{2}}$ and $v \in (-1,1)$.

The Lorentz boost of a static kink travels with velocity v and is subject to a phenomenon called Lorentz contraction. This is where the shape of a Lorentz-boosted kink is sharper than the static kink, and it happens because $\gamma > 1$ when $v \neq 0$, which scales the x values correspondingly.

Theorem 5.0.3. *[Energy and Momentum of Lorentz Boosted Field] Let u be a static kink solution with rest energy M, then the energy E and momentum P of the Lorentz boost of this kink $\tilde{u} = u(\gamma(x - vt))$ is as follows:*

$$E = \gamma M$$
$$P = \gamma v M$$

5.1 Energy and Momentum of Sine-Gordon Kink

We are now in a position to easily work out the energy and momentum of (4), the kink solution for the sine-Gordon equation that I introduced in Section 2.2.

First, notice that (4) is just the Lorentz boost of the static kink

$$u(x) = 4\tan^{-1}(e^x) \tag{11}$$

To be able to apply Theorem 5.0.3, we first need to show that (11) satisfies the conditions for the theorem. This amounts to showing that (11) satisfies equation (3) with the correct kink boundary behaviour, and that (11) is a static kink (I will do this by showing that $\frac{du}{dx} = \sqrt{2V(u)}$, which equates to showing it is a static kink, as shown in Section 4). Once we have done this, we can work out the energy M of (11), and then we can easily work out the energy and momentum of (4) using Theorem 5.1.

Let $u(x,t) = 4\tan^{-1}(e^x)$. Then,

$$u_x = \frac{4e^x}{1+e^{2x}},\ u_{xx} = \frac{4(e^x - e^{3x})}{(1+e^{2x})^2},\ u_t = u_{tt} = 0$$

and

$$V'(u) = \sin u = \frac{4(e^x - e^{3x})}{(1+e^{2x})^2}$$

Substituting these values into (3) we get:

$$u_{tt} - u_{xx} + \sin u = 0 - \frac{4(e^x - e^{3x})}{(1+e^{2x})^2} + \frac{4(e^x - e^{3x})}{(1+e^{2x})^2} = 0$$

So (11) is a solution of the sine-Gordon equation. As $x \to -\infty$, $u(x,t) \to 0$ and as $x \to \infty$, $u(x,t) \to 2\pi$, so (11) obeys the kink boundary behaviour for the sine-Gordon equation, so it is a kink solution.

Now I will show that (11) is a static kink, by showing that $\frac{du}{dx} = \sqrt{2V(u)}$. We know that $\frac{du}{dx} = u_x = \frac{4e^x}{1+e^{2x}}$, so we just need to work out $\sqrt{2V(u)}$. Using the trigonometric identity $1 - \cos(x) = 2\sin^2(\frac{x}{2})$, we get

$$V(u) = 1 - \cos\big(4\tan^{-1}(e^x)\big) = 2\sin^2(2\tan^{-1}(e^x))$$

so,

$$\sqrt{2V(u)} = 2\sin\big(2\tan^{-1}(e^x)\big) = \frac{4e^x}{1+e^{2x}}$$

Clearly, $\frac{du}{dx} = \sqrt{2V(u)}$, so $u(x,t)$ is a static kink. Hence, the conditions for Theorem 5.1 are satisfied, so if we can work out the energy M of (11), then we can easily work out the energy and momentum of its Lorentz boost (4). Since (11) does not depend on t, all of its t-derivatives equal 0, so we can leave out the u_t term in our equation for M.

$$\begin{aligned}
M &= \int_{-\infty}^{\infty} (\frac{1}{2}{u_x}^2 + V(u))dx \\
&= \int_{-\infty}^{\infty} \frac{8e^{2x}}{(e^{2x}+1)^2}dx + \int_{-\infty}^{\infty} \frac{8e^{2x}}{(e^{2x}+1)^2}dx \\
&= 2\int_{-\infty}^{\infty} \frac{4e^{2x}}{(e^{2x}+1)^2}dx + 2\int_{-\infty}^{\infty} \frac{4e^{2x}}{(e^{2x}+1)^2}dx \\
&= 2\int_{-\infty}^{\infty} \operatorname{sech}^2(x)dx + 2\int_{-\infty}^{\infty} \operatorname{sech}^2(x)dx \\
&= 2\tanh(x)\Big|_{-\infty}^{\infty} + 2\tanh(x)\Big|_{-\infty}^{\infty} \\
M &= 4 + 4 = 8
\end{aligned}$$

Now that we have found M, we can apply theorem 5.1 to find that the energy of the kink (4) is 8γ, and it's momentum is $8\gamma v$, where $\gamma = (1-v^2)^{-\frac{1}{2}}$.

5.2 Energy and Momentum of ϕ^4 Kink

Now, I wish to do the same to find the energy and momentum of the ϕ^4 kink solution (6).

Again, notice that (6) is the Lorentz boost of the following static kink

$$u(x) = \tanh(x) \tag{12}$$

First, I will show that (12) is a solution of (5). We have

$$\begin{aligned}
u_{tt} &= 0 \\
u_{xx} &= -2\tanh(x)\operatorname{sech}^2(x) \\
V'(u) &= -2u(1-u^2) = -2\tanh(x)\operatorname{sech}^2(x)
\end{aligned}$$

Substituting these into the ϕ^4 equation, and dividing by common terms, we arrive at $u_{tt} - u_{xx} + V'(u) = 2 - 2 = 0$, so (12) is a solution of the ϕ^4 model.

The vacua for the ϕ^4 model are $u_0 = -1$ and $u_1 = 1$. Clearly, $\tanh(x)$ tends to u_0 as $x \to -\infty$ and $\tanh(x)$ tends to u_1 as $x \to +\infty$. So, the solution (12) exhibits the correct kink boundary behaviour. So (12) is a kink solution of (5).

Now to show that (12) is static. We have

$$\frac{du}{dx} = \operatorname{sech}^2(x)$$

and

$$\sqrt{2V(u)} = 1 - \tanh^2(x) = \operatorname{sech}^2(x)$$

So, $\frac{du}{dx} = \sqrt{2V(u)}$, hence (12) is a static kink, of which (6) is the Lorentz boost. Now, if I can find the rest energy M of (12), then I can apply Theorem 5.0.3 to work out the energy and momentum of (6). Note the similarity to working out the Bogomol'nyi bound for a ϕ^4 kink.

$$\begin{aligned} M &= \int_{-\infty}^{\infty} (\frac{1}{2}{u_x}^2 + V(u))dx \\ &= \int_{-\infty}^{\infty} (\frac{1}{2}\operatorname{sech}^4(x) + \frac{1}{2}\operatorname{sech}^4(x))dx \\ &= \tanh(x) - \frac{1}{3}\tanh^3(x)\Big|_{-\infty}^{\infty} \\ &= (1 - \frac{1}{3}) - (-1 + \frac{1}{3}) = \frac{4}{3} \end{aligned}$$

So, $M = \frac{4}{3}$. Using Theorem 5.0.3, the energy of (6) is $E = \frac{4}{3}\gamma$ and the momentum is $P = \frac{4}{3}\gamma v$.

6 Kink-Antikink Collisions in the ϕ^4 Model

Sources used in this section are [1], [2], [4], [5], [8] and [9].

Solitons appear naturally in various physical systems. Understanding their interactions with each other helps us gain more insight on these naturally-occurring phenomena. Drawing on much of what we have covered in the previous sections, I will study their collisions with each other in more detail for the rest of this paper.

I have defined two different potentials in this paper which give rise to two different equations that admit kink soliton solutions - the sine-Gordon equation and the ϕ^4 equation. For the sine-Gordon equation, due to its integrable nature, we can find analytic multisoliton solutions through various integration methods and easily plot these for different times and velocities [5]. In the soliton collisions for this model, the solitons pass through each other with nothing more than a phase shift, keeping their shape and velocity [1] [5]. However, these fully elastic types of interactions are rarely seen in real life scenarios, and most of the models underpinning appearances of solitons are not analytically solvable to find multi-kink solutions. For example, if trying to understand the inelastic collisions of hadrons, the sine-Gordon equation does not act as a possible model for these interactions, due to the elastic properties of the sine-Gordon model that I have just outlined [5]. Commonly, there is some physical perturbation in the real life system that does not allow for complete integrability of the model, or the model itself is just non-integrable, such as the ϕ^4 model [1] [5]. In these cases, the interactions between waves is non-trivial, and the real world is more accurately modelled. This is evident for the ϕ^4 model when considering the frequent occurrences in a number of topics, as analysed in [1]. For this reason, I will be focusing on kink-antikink interactions in the ϕ^4 model for the rest of this paper.

6.1 The Numerical Method for Simulating Kink-Antikink Collisions

Again, my code can be found on GitHub: https://github.com/moritzstuebing/Kink-Antikink-Collisions-in-the-Phi-4-Model-with-a-Finite-Difference-Method

The ϕ^4 model does not admit any analytic multi-kink solutions, meaning there is no 'easy' way to plot our kink-antikink interactions for different times. Hence, I will be solving the model numerically, which I will do by writing a finite difference programme in MATLAB. Instead of thinking of our solution $u(x,t)$ as a continuous function, we will instead think of it as $u_{j,k}$ which takes values for discrete points in space-time (j,k). This involves discretising the space-time domain into a lattice, which we can think of as a matrix, and then using a finite difference scheme to find the values of our solution for each of the different space-time points (each entry in our matrix). To be able to do this, I need to find initial conditions for our first and second time 'slice', and then I can write a loop which will find the next 'slice', which I can then iterate, to create a matrix $U(j,k)$ where each column represents our field at a certain time. Once I have accurately written the code for my scheme, I will plot some of these columns (which correspond to the kink-antikink interaction for various times), to show how our collisions progress as time goes on. Then, I will experiment with different initial velocities to see what different phenomena I can observe.

6.1.1 Creating and Calibrating Spatial/Temporal Axis, and Generating Solution Matrix $U(j,k)$

I need to define a few variables and constants before I can move onto the main bulk of the method of my simulation. Firstly, I want to convert our position in space x into a vector. For my plot, after experimenting with different axis sizes, I picked the length of the x-axis to be $L = 150$ either side of the origin, so the full axis has a length of $2L = 300$, with $x = [-150, 150]$. When plotting the graphs, I limited the x-axis to $[-15, 15]$. I picked this to be the axis size as it gives enough space to see what happens in the full kink-antikink interaction. One important issue with a finite x grid is that any energy emitted as ripples during the collisions will travel to the boundaries of the axis, be reflected, and travel back to the center [1] [9]. This could cause this previously 'lost' energy to interact with the kink/antikink leading to behaviour that would not commonly occur. Previously, I used a much smaller axis size, and I noticed that, due to the reflection of these ripples, some of the behaviour I observed was not in coherence with many of the published findings in the papers I had researched. This is why I picked such a large axis size, so that the ripples did not have time to reflect and impact the behaviour of the kink-antikink pair.

Although our method does not give rise to a classically continuous solution - as we have discretised our domain - we want to give an impression as though our solution is continuous when plotting it. If the spatial step size h_x is too big, then the 'jumps' between points will be very obvious when plotting our solution, which is undesirable. I also didn't want the spatial step size h_x to be too small as this would increase the size of our matrix, leading to larger computation times and issues with memory capacity. After taking all of this into account, and experimenting with different spatial step sizes, I found the spatial step size of $h_x = 0.01$ to be an ideal balance between these issues. To arrive at this step size, I needed to work out the desired number of spatial steps N. This was a matter of easy algebra: I knew I wanted $h_x = \frac{2L}{N} = \frac{300}{N} = 0.01$, so I arrived at $N = 30000$.

Next, I needed to do the same for the time 'axis'. I tested out different time limits (i.e length of the time axis), and concluded that I wanted the time limit of $t = 200$. This allowed me to see the full interaction between the kink and antikink, but also was not so long that my solution matrix became too big and cumbersome for MATLAB to compute (i.e exceeded MATLAB's allowed array size). I may have been able to use a time limit that was slightly smaller, but I decided not to take the risk of missing out on interactions between solitons due to them happening after the final time slice.

Now to find the temporal step size h_t. The relationship between the size of the spatial steps and size of the temporal steps is very important for the stability and accuracy of our programme. In order for the scheme to work, it needed to obey the Courant-Friedrichs-Lewy condition of $|\frac{h_t}{h_x}| < 1$ [1] [5]. In other words, the h_t value needed to be less than h_x. Hence, I endeavoured to use a temporal step size of $h_t = 0.008$. Since my time limit is $tim = 200$, I knew that I needed the number of time steps $K = 25000$, so that $h_t = tim/K = 200/25000 = 0.008$.

Once I had an array of size N for the spatial axis and an array of size K for the temporal axis, I created a matrix $U(j,k)$ with size $N \times K$. This matrix will be used to store our solution; each $U(j,k)$ represents the value of our solution at the position (j,k) in our space-time lattice. Each row can be thought of as the evolution over time of one specific point on the spatial axis, and each column as our kink-antikink pair at that specific time step (i.e at time corresponding to $(k-1) \times h_t$, for the k'th column). This would allow me to plot the whole kink/antikink pair at any point in time, or the evolution of any one point over time (later I did this to plot $u(0,t)$). Once this was all set up, I was now ready to begin filling in my solution matrix U.

6.1.2 The Main Loop

Before we can insert our initial conditions into the matrix, we need to create the main loop of our finite difference scheme. This loop then takes the initial conditions and generates the solution for further times.

For any j from 1 to N, the position on our x-axis corresponding to this j will be $x = ((j-1) \times h_x) - L$, where L is the maximum distance of the x axis from the origin. For all values of k, the position in time corresponding to this k will be $t = (k-1) \times h_t$. I used this when plotting different time slices and different positions in x.

To fill in the solution matrix, I wanted to find an equation that uses the last two time slices in the matrix, and evaluates the next time slice using the previous two. This is what the finite difference scheme was able to do. To work out the value of a point $U(j,k+1)$, we needed the value of $U(j,k)$, $U(j-1,k)$, $U(j+1,k)$ and $U(j,k-1)$. If all of these values were available, then our scheme could work out the value of $U(j,k+1)$. Let me explain in more detail how I developed this scheme.

Our equation is a non-linear partial differential equation. Clearly, as it is a PDE, it involves partial derivatives. I needed to find a way to approximate these partial derivatives for each point in our space-time lattice. The way I did this was by using the Taylor series, which approximates partial derivatives with the following equations:

$$u_{xx} \approx \frac{u_{i+1,j} - 2u_{i,j} + u_{j-1,k}}{h_x^2}$$

$$u_{tt} \approx \frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{h_t^2}$$

Once I had these approximations, I considered the ϕ^4 equation:

$$u_{tt} - u_{xx} - 2u + 2u^3 = 0$$

Using $u(x,t) \to u_{j,k}$ (which we can think of in matrix form as $U(j,k)$), and plugging in my Taylor series approximations I arrived at the following equation for $u_{j,k+1}$:

$$u_{j,k+1} = h_t^2(\frac{u_{j+1,k} - 2u_{j,k} + u_{j-1,k}}{h_x^2}) - h_t^2(2u_{j,k}^3 - 2u_{j,k}) + 2u_{j,k} - u_{j,k-1}$$

This equation works out the $(j, k+1)$ value using only the $(j+1, k)$, (j, k), $(j-1, k)$ and $(j, k-1)$ values. Now, I had to iterate this equation in a loop to work out the values in the matrix $U(j, k)$. I did this by first working out each j value per time slice before moving on to the next time slice.

However, notice that if there are no previous two time slices, then this equation won't work. Also, if you are on the boundary of the matrix, then there is no value to the left/right of that point (depending which side of the matrix you are on). These are problems that I will address in the next section.

6.1.3 Boundary Values and Initial Conditions

As I said, if you are trying to work out the value of $U(1, 10)$, for example, then the equation requires you to know the value of $U(0, 9)$. Similarly, if trying to work out $U(N, 10)$, then you need the value $U(N+1, 9)$. These values simply don't exist because the j-index of the matrix only ranges from 1 to N. So, we needed to find a different way to handle these boundary values. As the leftmost and rightmost value of the kink-antikink pair is always -1, I decided to simply fill in the boundary values to be -1 for all time slices. In MATLAB matrix terms, this meant that $U(1, :) = U(N, :) = -1$. I then iterated the loop only for j values from 2 to $N-1$. This eliminated the boundary value problem of my loop equation requiring values outside of the matrix to work out values at $j = 1$ and $j = N$.

The loop requires two previous time slices to work out the next. But then this brings up the question of how to work out the very first two time slices. If I try to just work these out using my loop, it is going to get stuck as the loop needs values from time k and $k-1$ to work out the value at time $k+1$. So to work out a value at $k = 1$, it would need the value at $k = 0$ and at $k = -1$. The same holds for trying to work out the value at $k = 2$, the loop would require a value at $k = 0$. Again, these values simply don't exist, so I couldn't work out the first two time slices using the loop. Hence, I had to define these time slices individually.

The $U(:, 1)$ time slice (when $k = 1$) is where we set up the original static kink-antikink pair before 'shooting' them at each other, using the second time slice. I did this by defining a piece-wise function, that takes the value of the static kink, as covered in Section 4, for $x < 0$ and the static antikink for $x \geq 0$. Since I am working in the ϕ^4 model, the static kink solution is $u(x) = \tanh(x)$ and $u(x) = -\tanh(x)$ for the static antikink. So, I defined a piecewise function u_0 that took the value $\tanh(x - a)$ for $x < 0$ and $-\tanh(x + a)$ for $x \geq 0$, and assigned this function to my first time slice, i.e $U(:, 1) = u_0$. This defined the original kink-antikink pair at time $t = 0$.

The value of a defines the starting distance of each kink from the origin. The length of this distance has a profound effect on the behaviour of our collisions. For large a, the kinks have more time to gain energy while travelling towards each other, meaning they collide with more energy. This means that, for large a, less initial velocity is required for the kink-antikink pair to reflect or "bounce", as they gain more energy during their motion. The opposite holds for smaller a; as they generate less energy during their motion, they are more likely to enter an oscillatory state [9]. Both these reflections and oscillatory states will be covered in detail in the next subsection. After some research, I chose to use $a = 7$. This was close enough to not warrant a large amount of time to reach the first interaction, and large enough so that the kink-antikink pair don't just immediately interact with each other, which would restrict the behaviour we could observe.

Now I needed to 'shoot' the kink-antikink pair at each other. I did this by defining a velocity field on the next time slice $U(:,2)$. Taking the derivative of the Lorentz boost (covered in Section 5) of u_0, and labelling this u_1, I defined the next time slice with $U(:,2) = u_0 + h_t u_1$. This allowed me to fire the kink/antikink at each other with initial velocity v_i, and I could vary v_i as desired. Notice that I needed to take the derivative of the Lorentz boost of the static kink-antikink pair, not simply just the static u_0, so that I could give my kink-antikink pair an initial velocity. Otherwise, the pair would just not move.

Once I had my first two time slices (the initial conditions), I could simply use my loop to iterate in space from $j = 2$ to $j = N - 1$, for each time step $k = 2$ to $k = K - 1$, and fill in my matrix. To observe the evolution of my kink-antikink pair for varying initial velocities, I could then simply plot the kink-antikink pair for specific times, by plotting x against the column of the matrix corresponding to the desired time, or the progression of a single point over time, by plotting one of the rows against t. This was done simply using the “plot” function in MATLAB’s command window.

6.2 The Kink-Antikink Collisions for Varying Initial Velocities

Now that I have set up the scheme that will allow me to simulate kink-antikink collisions in the ϕ^4 model, I will see what different phenomena I can observe when varying the initial velocity v_i of the kink-antikink pair.

6.2.1 Initial Velocities Greater Than v_c and Less Than v_l

One key thing that has been observed in prior research of the topic, is that there is a critical velocity v_c upon which much of the observed behaviour depends on. For all initial velocities $v_i > v_c$, the kink and antikink always reflect inelastically. Moshir [5] gives v_c to be $0.2594 +/- 0.0001$, Wingate [9] gives it to be 0.26, and Campbell and Peyrard [2] give it to be 0.3. Other papers I have researched, for example [8], also give slightly varying values for v_c. The important point here is that there is some threshold for the initial velocity above which only reflection occurs. In coherence with [1], I could not observe any oscillation (non-reflection) cases for $v_i > 0.2598$, so I will take this to be v_c.

As mentioned, for initial velocities greater than $v_c = 0.2598$, the kink-antikink pair always reflect inelastically. In fact, in the ϕ^4 model all reflection/scattering cases are inelastic [1] [2]. When I mention that they reflect inelastically, what I mean is that the final velocity v_f of the kink-antikink pair is less than the initial velocity v_i. This suggests that some energy has been lost during the collision. Also, when observing this reflection of the kinks, it may look like the pair has passed through each other, but it is important to note that, in the ϕ^4 model, the kink and the antikink will never pass through each other, unlike in the sine-Gordon equation [9].

Figures 5 and 6 show the evolution of a kink-antikink pair that have been fired at each other with initial velocity $v_i = 0.35 > v_c$. As you can see, the kink and the antikink collide in the middle and reflect off each other, before travelling off to infinity. Also notice that when the pair collide, there is a dip of the graph to ≈ -1.7, before the pair bounce back up and travel to infinity. All of these results are accurate when compared to the results from previous papers ([1] [2] [5] [9]).

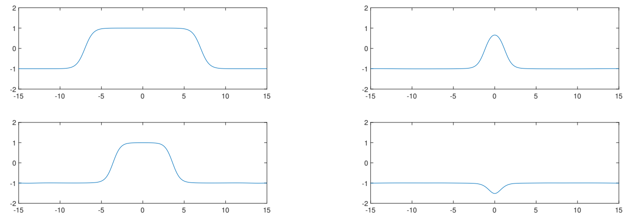

Figure 5: The kink-antikink pair for initial velocity $v = 0.35$, at times $t = 0$ (top left), $t = 10$ (bottom left), $t = 17$ (top right), $t = 19$ (bottom right)

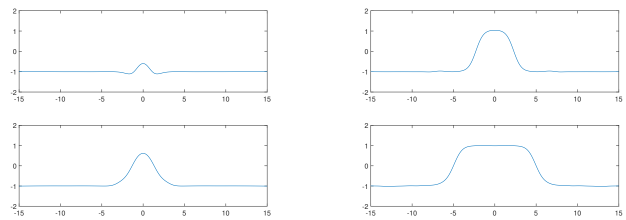

Figure 6: The kink-antikink pair for initial velocity $v = 0.35$, at times $t = 20$ (top left), $t = 22$ (bottom left), $t = 27$ (top right), $t = 45$ (bottom right)

An interaction like this, where the kinks collide once and then reflect off to infinity, is called a "one-bounce" interaction. We say "bounce" because of the way the kink-antikink pair looks like they bounce of each other. Later on, I will study some collisions where more than one bounce is observed.

Figures 5 and 6 shed some light on the behaviour of kink-antikink collisions for initial velocities greater than the critical velocity v_c. Below v_c, the phenomena observed is more complicated. There is another important threshold for the initial velocity beneath which the behaviour of the collisions is ubiquitous. I will call this velocity $v_l \approx 0.193$ (I first found this value in [1] and then ran some tests to see that my simulation agrees with this value, hence I am also using this value). For $v_i < v_l$, a trapped state for the kink-antikink pair is formed, where the pair is bound together by their mutual attraction into a persistent oscillatory state. This state only occurs if the pair are travelling at a low enough velocity to have enough time to lose sufficient energy in the collision [1]. Figures 7 and 8 show the kink-antikink collision for an initial velocity $v = 0.1$. The oscillation between $u = 1$ and $u \approx -1.7$ can be seen.

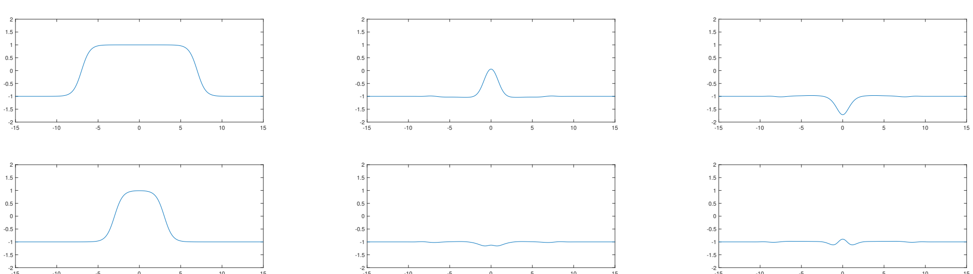

Figure 7: The kink-antikink pair for initial velocity $v = 0.1$, at times $t = 0$ (top left), $t = 40$ (bottom left), $t = 65.2$ (top centre), $t = 66$ (bottom centre), $t = 66.4$ (top right) and $t = 67$ (bottom right)

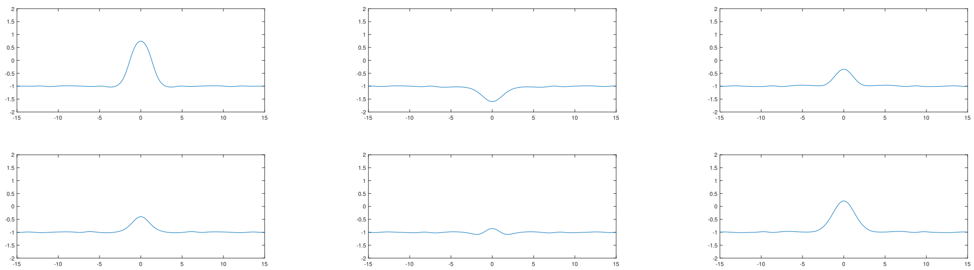

Figure 8: The kink-antikink pair for initial velocity $v = 0.1$, at times $t = 71$ (top left), $t = 74$ (bottom left), $t = 75$ (top centre), $t = 75.6$ (bottom centre), $t = 76$ (top right) and $t = 76.8$ (bottom right)

So far, I have observed the behaviour of our kink-antikink collisions for initial velocities greater than $v_c = 0.2598$, and for initial velocities less than $v_l = 0.193$. Above v_c, we find that our kink-antikink pair always reflect inelastically, meaning they lose some energy in the collision and their final velocity is less than their initial velocity. At initial velocities lower than v_l, we found that a permanent oscillatory state is created, since the low velocity gives the kinks enough time to lose sufficient energy to enter this trapped space after the collision [1].

6.2.2 Initial Velocities Between v_c and v_l

I now wish to find out what happens for initial velocities between v_c and v_l. Sugiyama [8] found that, in all cases below v_c, the kink and antikink are trapped in a bound state. Moshir [5] observed two narrow regions in this range $v_l < v_i < v_c$ where scattering occurs. As time has gone on, more and more regions of reflection have been found for different v_i in this intermediate region. Campbell, Schonfeld and Wingate [1] found more than two alternating regions of trapping and reflection. They observed 9 scattering regions, alternated with regions of trapped oscillation.

The trapped oscillation cases are very similar to the case I showed earlier, but the reflection cases see some difference. As mentioned before, for initial velocities greater than v_c, we see a one bounce interaction; the kink-antikink pair collides in the middle and then reflects back to infinity inelastically. Interestingly, for intermediate v_i, the reflection windows I observed were "two bounce" windows. This means that after the original collision, the kink-antikink pair reflect to a finite distance apart, before changing direction and colliding again, leading to a second reflection, after which they recede to infinity [1]. Moshir [5] observed this in his 1981 paper for velocity $v_i = 0.224$, but at the time the terminology of "bounces" had not been coined.

As I have already presented one of the oscillatory states, and not much will be different, in the intermediate region, from the case I have shown, I will now focus on presenting these two bounce interactions. Figures 9, 10 and 11 present one case at the lower end of the intermediate spectrum, for initial velocity $v_i = 0.198$. The original reflection, separation to a finite distance, and then re-collision and reflection off to infinity can all be seen in these figures as t increases.

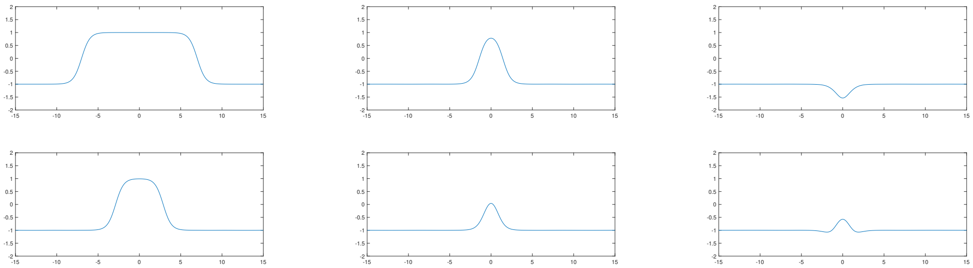

Figure 9: The kink-antikink pair for initial velocity $v = 0.198$, at times $t = 0$ (top left), $t = 21$ (bottom left), $t = 28$ (top centre), $t = 30$ (bottom centre), $t = 31$ (top right) and $t = 32$ (bottom right)

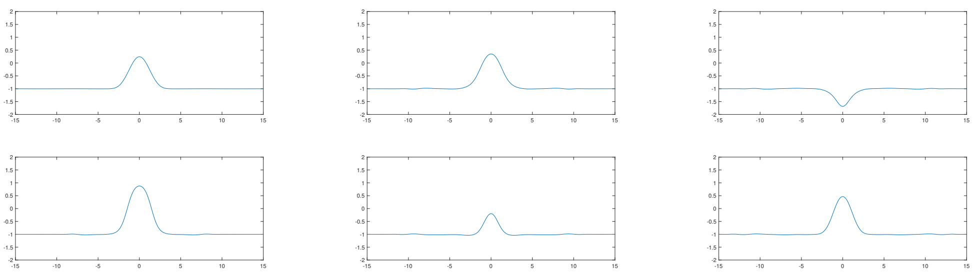

Figure 10: The kink-antikink pair for initial velocity $v = 0.198$, at times $t = 33$ (top left), $t = 40$ (bottom left), $t = 42$ (top centre), $t = 43$ (bottom centre), $t = 44$ (top right) and $t = 46$ (bottom right)

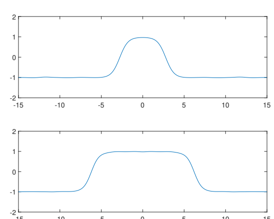

Figure 11: The kink-antikink pair for initial velocity $v = 0.198$, at times $t = 55$ (top) and $t = 77$ (bottom)

In an effort to save space, I will not show the full kink-antikink interaction for any other intermediate velocities in the range between v_l and v_c. Instead, I will show the graphs of $u(0, t)$ for some of these to give an idea of the the bounce behaviour.

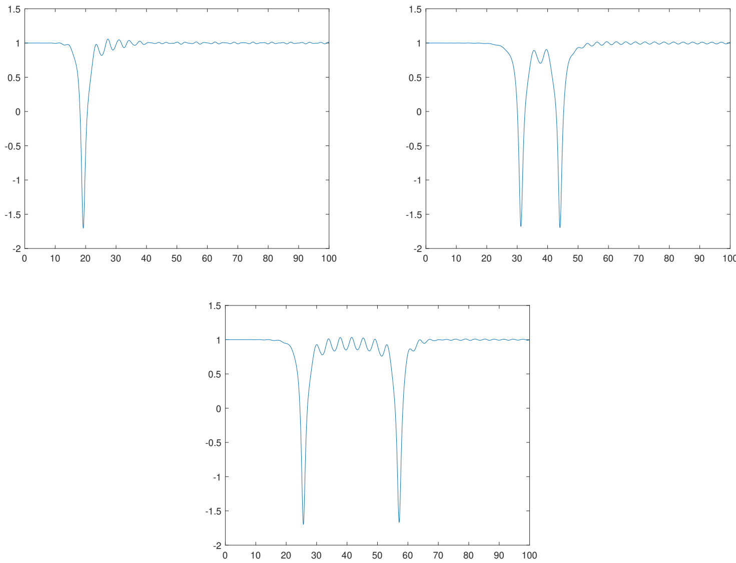

Figure 12: Graph of $u(0,t)$ for initial velocity $v_i = 0.35$ (top left) $v_i = 0.198$ (top right) and $v_i = 0.24855$ (bottom), up to time $t = 100$. The dips to ≈ -1.7 shows the points where the kink-antikink pair collide and bounce off each other.

Figure 12 shows $u(0,t)$ for various different initial velocities. The top left plot show $u(0,t)$ for $v_i = 0.35 > v_c$. The dip in the graph to ≈ -1.7 occurs at the moment when the kink and antikink collide in the middle and bounce off each other. In comparison with the other two plots, which both correspond to initial velocities between v_l and v_c, we can see that it only has one dip, while the others have two. This shows the difference in bounce behaviour: one dip indicates one bounce, and two dips indicate two bounces. When analysing these interactions, we notice two things. Firstly, the initial collision occurs earlier for the higher velocity. This is intuitive, as a higher initial velocity causes the kink/antikink to reach each other sooner. As we can see, the earliest interaction occurs for $v_i = 0.35$ and the latest for $v_i = 0.198$. Secondly, for the two bounce interactions, the time between the bounces is larger for higher velocities. This makes sense, as they initially collide with more energy, so they have more energy to separate after the first bounce, meaning they travel further distance apart after the first bounce, leading to more time before they return to collide again.

In this section, I have studied kink-antikink collisions for various initial velocities v_i. I have observed that for $v_i > v_c = 0.2598$, the kink-antikink pair always collide at the origin, and then reflect inelastically. I have shown this for initial velocity $v_i = 0.35$. Next, I studied another threshold for the initial velocity, $v_l = 0.193$, under which we always observe the kink and antikink being trapped in a permanent oscillatory state. I have plotted this interaction-type for $v_i = 0.1$. Finally, I observed some of the behaviour for initial velocities between v_l and v_c, finding that there are alternate regions of trapping and reflection for initial velocities in this range. I have plotted one of the reflection cases for $v_i = 0.198$, and then compared $u(0,t)$ for $v_i = 0.35$, $v_i = 0.198$, $v_i = 0.24855$ to observe that, for initial velocities in the intermediate region, the kink-antikink pair form a two bounce interaction, where they reflect twice. I have studied some of the key differences between the two bounce interactions.

7 Conclusion

In this report, we have gained a deeper understanding of kink solitons and their collisions in the ϕ^4 model. Two different equations that admit kink solutions have been looked at: the sine-Gordon equation and the ϕ^4 equation. The latter has been investigated in more detail, as the kink/antikink collisions were studied in the ϕ^4 model. Different solutions to these equations have been presented and verified, and the conservation laws for these solutions were introduced. The energy density of the ϕ^4 model was calculated and plotted for various times and velocities. Next, static kinks and the Bogomol'nyi argument were researched, and this was given for the ϕ^4 model to find a bound on the energy of a ϕ^4 kink. The Lorentz boost was next up, which was used to find the energy and momentum of a ϕ^4 kink, by applying a theorem which made this task simpler. In the final section, kink-antikink collisions in the ϕ^4 model were the focus. First, the numerical method - a finite difference scheme - that was used to simulate the collisions was explained, as well as the reasoning behind choices for parameters and variables. Finally, observations were presented for what happens at different initial velocities. The interactions were plotted and any interesting behaviour was discussed.

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